

# **Companion to Bird & de Moor, Algebra of Programming**

## §8. Thinning Algorithms

### §8.1. Thinning

(8.1)

For  $Q : A \rightarrow A$ ,  $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : PA \rightarrow PA$  (8.1).

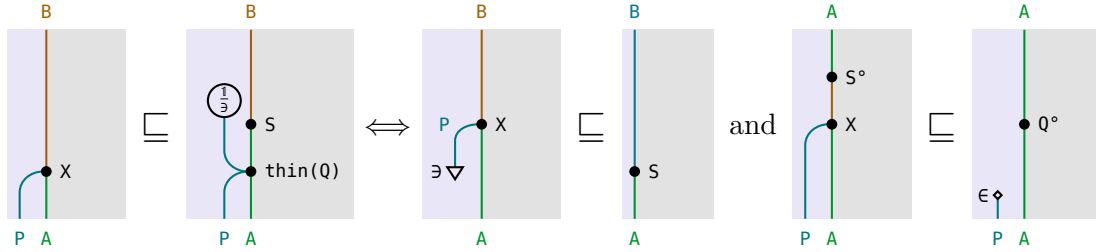
$ys \text{ thin}(Q) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$  the same  $^\circ$  as  $\triangleq \text{est}$ :  $\epsilon X$  runs  $a \rightarrow ys \rightarrow xs$ , member of  $ys$  first, and  $Q^\circ \epsilon$  runs  $a \rightarrow b \rightarrow xs$ , so  $(a, b) \in Q^\circ$  reads  $b \ Q \ a$ ; at  $Q^\circ \leq$  every  $a \in ys$  keeps some  $b \leq a$  in  $xs$ , the end  $\text{est}(\leq)$  picks

(8.1a)

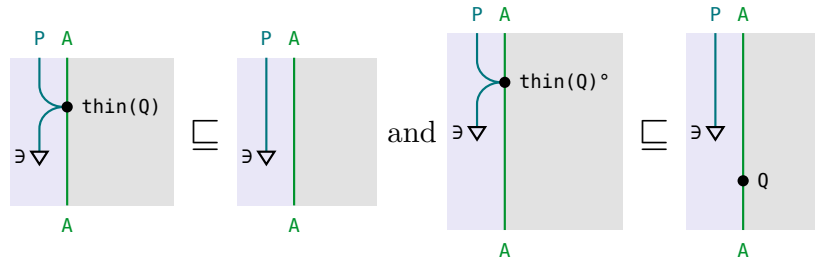
| the law                                                                                                                                                                                                                                                                        | what it says                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \exists S$ and $S^\circ X \subseteq Q^\circ \epsilon$                                                                                                                                                               | $S$ is the whole algebra: it produces every candidate that $\text{thin}(Q)$ thins. Everything kept is an $S$ -value, and every $S$ -value has a $Q$ -lower bound among the kept ones |
| 2 $\text{thin}(Q) \ni \exists \ni$                                                                                                                                                                                                                                             | everything a thinning keeps was in the set                                                                                                                                           |
| 3 $\text{thin}(Q)^\circ \ni \exists \ni Q$                                                                                                                                                                                                                                     | every element of the set has a $Q$ -lower bound among the kept ones                                                                                                                  |
| 4 $Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$                                                                                                                                                                                                             | the fewer pairs $Q$ relates, the fewer subsets count as thinnings                                                                                                                    |
| 5 $1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$<br>$Q \subseteq Q \implies \text{thin}(Q) \text{ thin}(Q) \subseteq \text{thin}(Q)$                                                                                                                                      | keeping everything is always a legal thinning                                                                                                                                        |
| 6 $Q \subseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \implies \text{est}(R) = \text{thin}(Q) \text{ est}(R) \quad Q \subseteq R, \text{ both preorders} \text{ — (8.1.2a) } \text{preorder}(Q) \implies \text{est}(R) \subseteq \text{thin}(Q) \text{ est}(R)$ | <b>thin-introduction</b> : thinning first cannot lose an $R$ -minimum                                                                                                                |
| 7 $\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists} \text{ (8.2) — (8.2) } \text{est}(Q) \frac{1}{\exists} \subseteq \text{thin}(Q)$                                                                                                                                   | <b>thin-elimination</b> : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on                                                                |
| 8 $\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \text{ (8.3), } R \cap (S^\circ S) \subseteq Q \text{ — thin-elimination variant}$                                                                                             | the usable variant: $R$ need only refine $Q$ between values $S$ gives one argument                                                                                                   |
| 9 $P(\text{thin}(Q)) \text{ union} \subseteq \text{union thin}(Q) \text{ (8.4)}$                                                                                                                                                                                               | thinning each member set is a thinning of the union                                                                                                                                  |

#### §8.1.1. $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \exists S \wedge S^\circ X \subseteq Q^\circ \epsilon$ — the definition of $\frac{S}{\exists} \text{thin}(Q)$

(8.1.1a)



(8.1.1b)



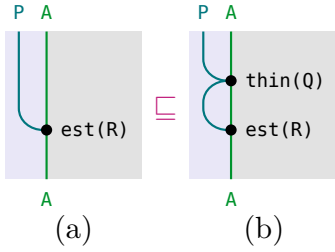
(8.1.1a) at  $X \triangleq \text{thin}(Q)$ ,  $S \triangleq \exists$ : the left side is  $\text{thin}(Q) \subseteq \text{thin}(Q)$ , since  $\frac{\exists}{\exists} = \wedge(\exists) = 1$

#### §8.1.2. $Q \subseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \implies \text{est}(R) = \text{thin}(Q) \text{ est}(R)$ — thinning first, then taking the smallest, is the same as taking the smallest

$$\text{preorder}(Q) \Rightarrow \text{est}(R) \sqsubseteq \text{thin}(Q) \quad (8.1.2a)$$

**est(R)**

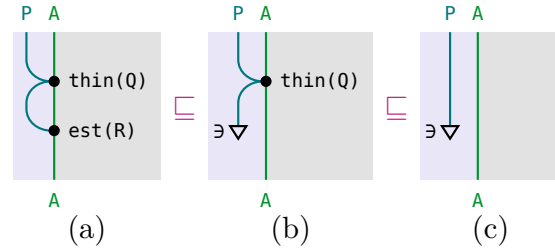
the  $\sqsubseteq$  half: keeping everything is a  
thinning —  $1 \sqsubseteq Q$



|     |                                             |                                                                                                                                                                                                                   |
|-----|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | est(R)                                      | $1 \sqsubseteq \text{thin}(Q) \text{ — } 1 \sqsubseteq Q$<br>$\Rightarrow 1 \sqsubseteq \text{thin}(Q),$<br>$Q \sqsubseteq Q$<br>$\Rightarrow \text{thin}(Q)$<br>$\text{thin}(Q) \sqsubseteq$<br>$\text{thin}(Q)$ |
| (b) | $\sqsubseteq \text{thin}(Q) \text{ est}(R)$ |                                                                                                                                                                                                                   |

$$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists \quad (8.1.2b)$$

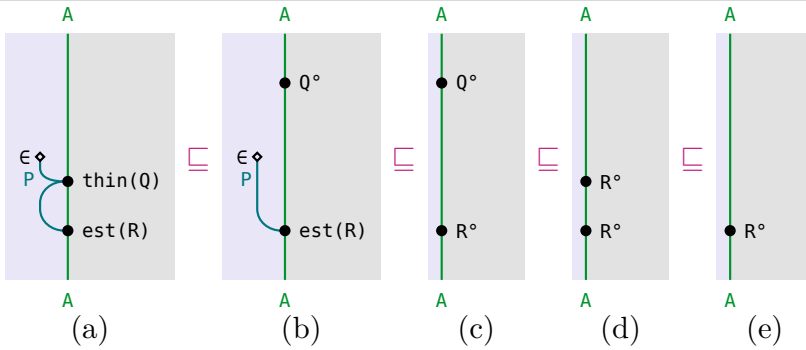
the  $\sqsubseteq$  half, first condition of the UP of  
est at  $X \triangle \text{thin}(Q) \text{ est}(R)$  — (7.1.1a)



|     |                                      |                                              |
|-----|--------------------------------------|----------------------------------------------|
| (a) | thin(Q) est(R)                       | $R \cap S \sqsubseteq R$                     |
| (b) | $\sqsubseteq \text{thin}(Q) \exists$ | $\text{thin}(Q) \exists \sqsubseteq \exists$ |
| (c) | $\sqsubseteq \exists$                |                                              |

$$Q \sqsubseteq R \wedge \text{preorder}(R) \Rightarrow \epsilon \text{ thin}(Q) \text{ est}(R) \sqsubseteq R^\circ \quad (8.1.2c)$$

the  $\sqsubseteq$  half, second condition —  $Q \sqsubseteq R, R$  a preorder



|     |                                               |                                                         |
|-----|-----------------------------------------------|---------------------------------------------------------|
| (a) | $\epsilon \text{ thin}(Q) \text{ est}(R)$     | $\epsilon \text{ thin}(Q) \sqsubseteq Q^\circ \epsilon$ |
| (b) | $\sqsubseteq Q^\circ \epsilon \text{ est}(R)$ | $\epsilon \text{ est}(R) \sqsubseteq R^\circ$           |
| (c) | $\sqsubseteq Q^\circ R^\circ$                 | $Q \sqsubseteq R$                                       |
| (d) | $\sqsubseteq R^\circ R^\circ$                 |                                                         |
| (e) | $\sqsubseteq R^\circ$                         |                                                         |

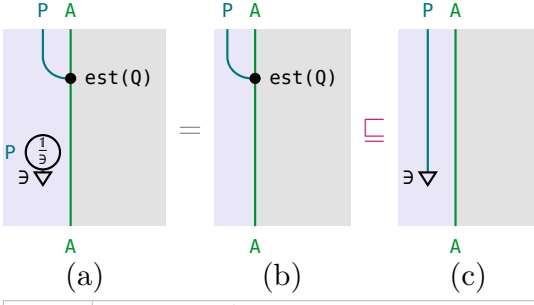
$$\mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq \mathbf{thin}(Q)$$

(8.2)

the singleton holding a  $Q$ -least member of a set is a thinning of that set

$$\mathbf{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

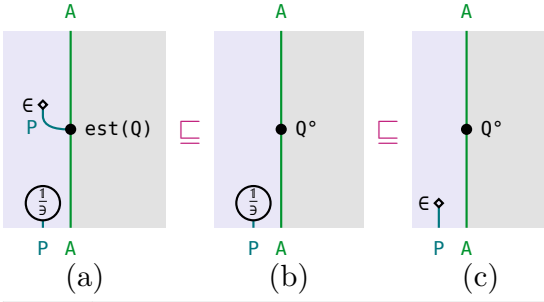
every member of the singleton is a member of the set —  $\triangleq \mathbf{thin}, \ni/\ni$  half



|     |                                             |                                 |
|-----|---------------------------------------------|---------------------------------|
| (a) | $\mathbf{est}(Q) \stackrel{1}{\exists} \ni$ | $\stackrel{1}{\exists} \ni = 1$ |
| (b) | $= \mathbf{est}(Q)$                         | $R \cap S \sqsubseteq R$        |
| (c) | $\sqsubseteq \ni$                           |                                 |

$$\in \mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \in$$

the singleton's member stands in  $Q$  to every member of the set —  $\triangleq \mathbf{thin}, \in/(Q^\circ \in)$  half



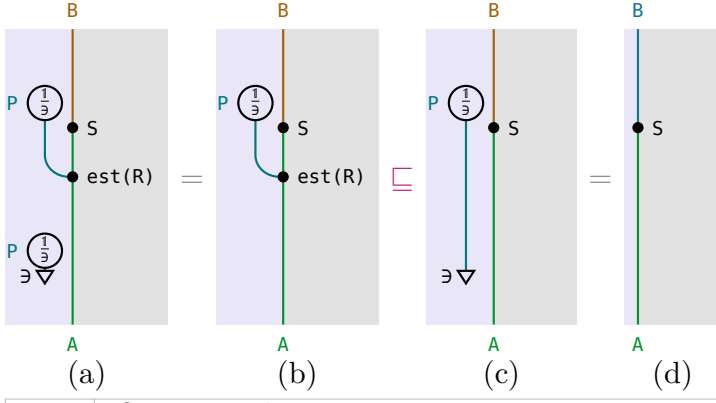
|     |                                             |                                           |
|-----|---------------------------------------------|-------------------------------------------|
| (a) | $\in \mathbf{est}(Q) \stackrel{1}{\exists}$ | $\in \mathbf{est}(R) \sqsubseteq R^\circ$ |
| (b) | $\sqsubseteq Q^\circ \stackrel{1}{\exists}$ | $\stackrel{1}{\exists} \sqsubseteq \in$   |
| (c) | $\sqsubseteq Q^\circ \in$                   |                                           |

$$R \cap (S^\circ S) \subseteq Q \Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq \frac{S}{\exists} \text{thin}(Q) \quad (8.3)$$

given  $R \cap (S^\circ S) \subseteq Q$ ,  $Q$  a preorder

$$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni \subseteq S$$

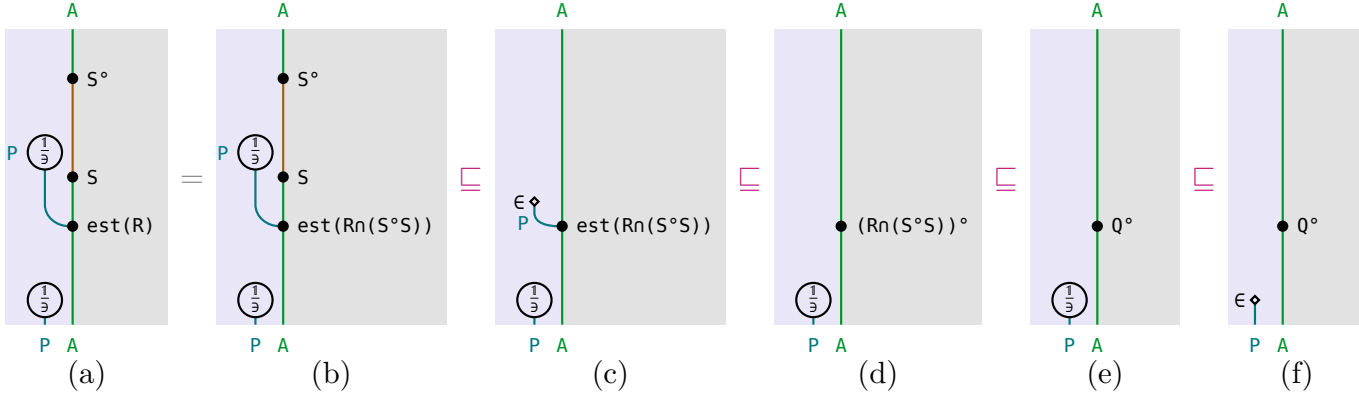
every member of the singleton is a value  $S$  returns — (8.1a.1), first condition



|     |                                                         |                             |
|-----|---------------------------------------------------------|-----------------------------|
| (a) | $\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni$ | $\frac{1}{\exists} \ni = 1$ |
| (b) | $= \frac{S}{\exists} \text{est}(R)$                     | $R \cap S \subseteq R$      |
| (c) | $\sqsubseteq \frac{S}{\exists} \ni$                     | $\frac{R}{\exists} \ni = R$ |
| (d) | $= S$                                                   |                             |

$$R \cap (S^\circ S) \subseteq Q \Rightarrow S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq Q^\circ \in$$

the singleton's member stands in  $Q$  to every value  $S$  returns at the same argument — (8.1a.1), second condition



|     |                                                                                |                                                                                      |
|-----|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| (a) | $S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$                    | $\frac{S}{\exists} \text{est}(R \cap (S^\circ S)) = \frac{S}{\exists} \text{est}(R)$ |
| (b) | $= S^\circ \frac{S}{\exists} \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$ | $T^\circ \frac{1}{\exists} \subseteq \in$                                            |
| (c) | $\sqsubseteq \in \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$             | $\in \text{est}(R) \subseteq R^\circ$                                                |
| (d) | $\sqsubseteq (R \cap (S^\circ S))^\circ \frac{1}{\exists}$                     | $R \cap (S^\circ S) \subseteq Q$                                                     |
| (e) | $\sqsubseteq Q^\circ \frac{1}{\exists}$                                        | $\frac{1}{\exists} \subseteq \in$                                                    |
| (f) | $\sqsubseteq Q^\circ \in$                                                      |                                                                                      |

$$\text{preorder}(Q) \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$$

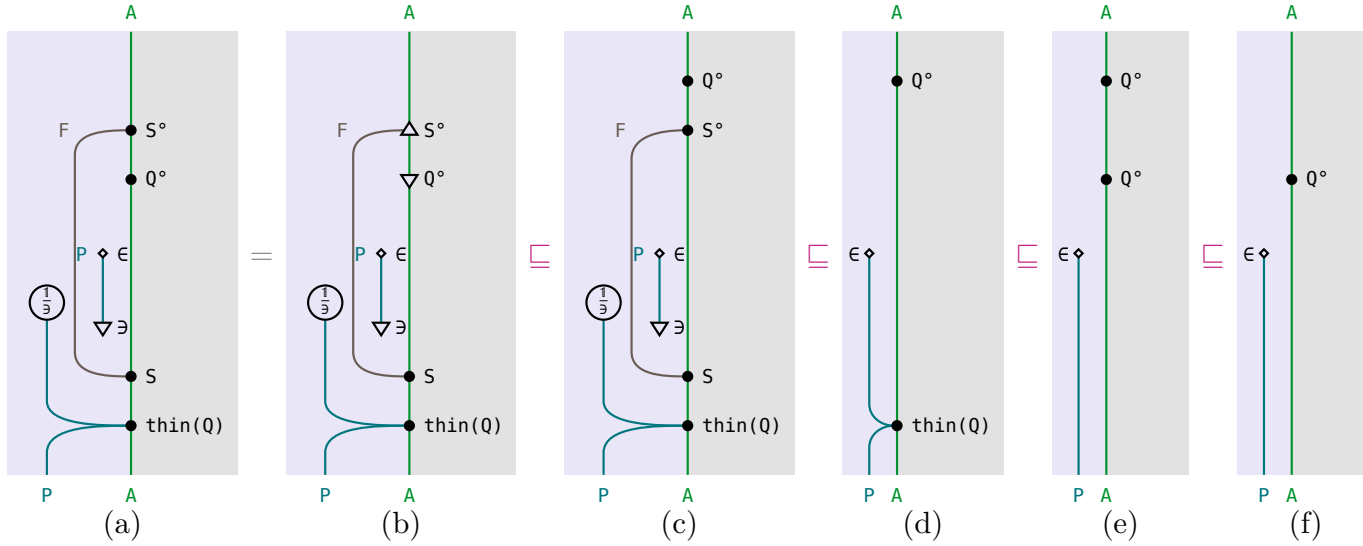
Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set —  $S$  monotonic on  $Q$ ,  $Q$  a preorder

$$\langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle (S) \rangle \quad \text{and} \quad \langle (S) \rangle \circ \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq Q^\circ \in$$

(8.1a.1) at  $X \triangleq \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle$ ,  $\langle (S) \rangle$  for its  $S$

$\Rightarrow$  fusion, hylomorphism theorem

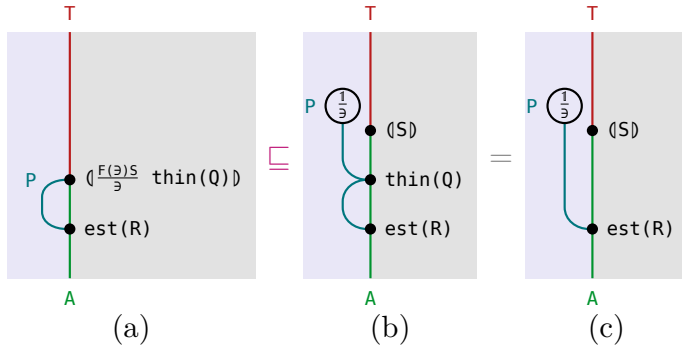


|     |                                                                                  |                                                               |
|-----|----------------------------------------------------------------------------------|---------------------------------------------------------------|
| (a) | $S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$             |                                                               |
| (b) | $= S^\circ F(Q^\circ) F(\in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$        | $\text{MonoAlg}(S, Q)$                                        |
| (c) | $\sqsubseteq Q^\circ S^\circ F(\in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$ | $\Rightarrow S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$  |
| (d) | $\sqsubseteq Q^\circ \in \text{thin}(Q)$                                         | $S^\circ F(\in) \frac{F(\exists) S}{\exists} \sqsubseteq \in$ |
| (e) | $\sqsubseteq Q^\circ Q^\circ \in$                                                | $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$                  |
| (f) | $\sqsubseteq Q^\circ \in$                                                        |                                                               |

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R)$$

Corollary 8.1

the thinning fold refines the optimisation problem itself —  $S$  monotonic on  $Q$ ,  $Q \sqsubseteq R$ , both preorders



|     |                                                                                |                                                                                                                                          |
|-----|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R)$    | Theorem 8.1 $\text{preorder}(Q) \wedge \text{MonoAlg}(S, Q)$                                                                             |
| (b) | $\sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R)$ | $\Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$ |
| (c) | $= \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R)$           | $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R)$                                                                    |
|     |                                                                                | $\Rightarrow \text{est}(R) = \text{thin}(Q) \text{est}(R)$                                                                               |

## §8.2. Paths in a layered network

(8.2a)

|    | name                            | type                                                               | definition                                                                             | meaning                                                                                                                                                          |
|----|---------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | F                               | $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$           | $F(B, X) = B + (B \times X)$                                                           | a path is either one vertex $a$ , or a vertex $a$ followed by a path $xs$                                                                                        |
| 2  | $\alpha$                        | $F(A, LA) \rightarrow LA$                                          | $\alpha = [\text{wrap}, \text{cons}]$                                                  | build a path: wrap one vertex as $[a]$ , or put $a$ in front of $xs$                                                                                             |
| 3  | wrapz                           | $A \rightarrow LA \times \mathbb{N}$                               | $\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle, a \mapsto ([a], 0)$ | a path of one vertex, no edge, cost 0                                                                                                                            |
| 4  | consw                           | $A \times (LA \times \mathbb{N}) \rightarrow LA \times \mathbb{N}$ | $(a, (xs, n)) \mapsto ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$                  | put $a$ in front of the path; the cost adds the weight of the new edge $a \rightarrow \text{head}(xs)$                                                           |
| 5  | cost                            | $LA \rightarrow \mathbb{N}$                                        | $\text{cost} \triangleq \llbracket [\text{wrapz}, \text{consw}] \rrbracket \pi_2$      | the cost of $[a_0, \dots, a_n]$ is $\text{wt}(a_0, a_1) + \dots + \text{wt}(a_{n-1}, a_n)$                                                                       |
| 6  | R                               | $LA \rightarrow LA$                                                | $R \triangleq \text{cost} \leq \text{cost}^\circ$                                      | $xs$ is cheaper than $ys$                                                                                                                                        |
| 7  | L(R)                            | $LA \rightarrow LB$                                                | $\llbracket F(\exists, 1) \alpha \rrbracket = L(\exists)$                              | $L(\exists)$ relates a list of layers to every path that chooses one vertex from each layer                                                                      |
| 8  | minpath                         | $L(PA) \rightarrow LA$                                             | $\text{minpath} \sqsubseteq \frac{L(\exists)}{\exists} \text{est}(R)$                  | the input $[x_0, \dots, x_n] : L(PA)$ is a list of layers, each a set of vertices; a path chooses one vertex from each layer, and minpath returns a cheapest one |
| 9  | Q                               | $LA \rightarrow LA$                                                | $Q \triangleq R \cap (\text{head} \text{ head}^\circ)$                                 | like $R$ , and the two paths also start at the same vertex: $\text{head}(xs) = \text{head}(ys)$                                                                  |
| 10 | l                               | $B \rightarrow B + C$                                              | $a \mapsto l(a)$                                                                       | mark a value as the left alternative                                                                                                                             |
| 11 | r                               | $C \rightarrow B + C$                                              | $b \mapsto r(b)$                                                                       | mark a value as the right alternative                                                                                                                            |
| 12 | cpl                             | $PB \times A \rightarrow P(B \times A)$                            | $\text{cpl} \triangleq \text{cp}(1 \times A)$                                          | pair every element of the set with the value beside it                                                                                                           |
| 13 | cpr                             | $A \times PB \rightarrow P(A \times B)$                            | $\text{cpr} \triangleq \text{cp}(A \times 1)$                                          | pair the value beside the set with every element of the set                                                                                                      |
| 14 | $\frac{F(\exists, 1)}{\exists}$ | $F(PB, X) \rightarrow PF(B, X)$                                    | $\frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$                             | cpl transpose: from layer $x$ and a set of paths $ps$ , choose a vertex $a \in x$                                                                                |
| 15 | $\frac{F(1, \exists)}{\exists}$ | $F(B, PX) \rightarrow PF(B, X)$                                    | $\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} P(l), \text{cpr } P(r)]$           | cpr transpose: from a vertex $a$ and a set of paths $ps$ , choose a tail $xs \in ps$                                                                             |
| 16 | step                            | $A \times P(LA) \rightarrow LA$                                    | $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$                      | put $a$ in front of every path in $ps$ and keep a cheapest one                                                                                                   |

(8.2b)

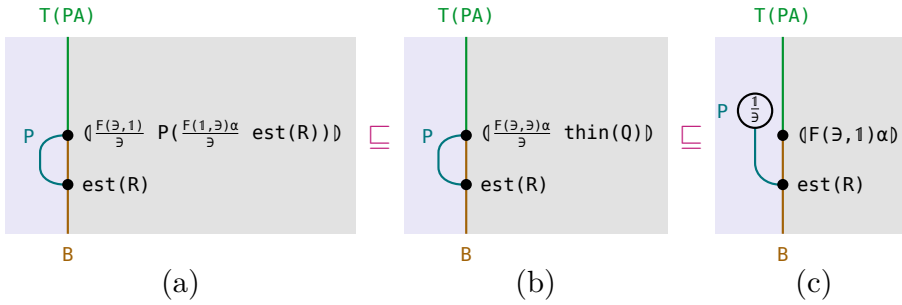
|   | the law                                                                         | what it says                                                                                                                                                             |
|---|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$                       | $F(\exists, 1) \alpha$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                                             |
| 2 | $S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$ | $S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \sqsubseteq Q$ : between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge \text{MonoAlg}(F(\exists, 1) \alpha, Q) \wedge R \sqcap ((F(1, \exists) \alpha)^\circ F(1, \exists) \alpha) \sqsubseteq Q \Rightarrow \quad (8.2c)$$

$$\left( \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \right) \text{est}(R) \sqsubseteq \left( \frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers

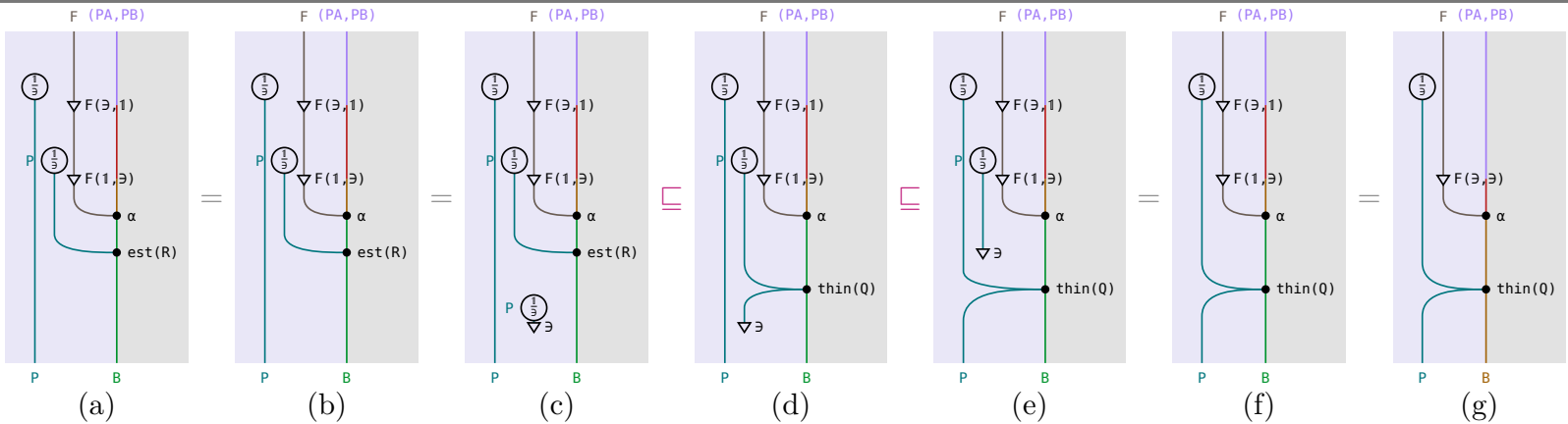
- Why it is needed:  $\text{est}(R)$  keeps one cheapest  $w$  of the set  $\Lambda(S)(x)$ ;  $\text{thin}(Q)$  needs every dropped  $z$  to be  $Q$ -beaten by a kept one. For  $\{w\}$  alone to be a thinning we need  $w \sqsubseteq z$  for every  $z$  in  $\Lambda(S)(x)$ .
- We have  $w \sqsubseteq z$  ( $w$  is cheapest) and  $w \sqsubseteq (S^\circ S) z$  (both are built by  $S$  from the same argument  $x$ ); the hypothesis turns the two into  $w \sqsubseteq z$ : among candidates from one argument, cheaper already means  $Q$ -better.



|     |                                                                                                                                  |                                                                                                                                                                                                                                      |
|-----|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\left( \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \right) \text{est}(R)$ | (8.2d) $R \sqcap (S^\circ S) \sqsubseteq Q$<br>$\Rightarrow \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$ |
| (b) | $\sqsubseteq \left( \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \right) \text{est}(R)$                             | $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge$<br>$\text{MonoAlg}(F(\exists, 1) \alpha, Q)$                                                                                                            |
| (c) | $\sqsubseteq \left( \frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$                                                  | $\Rightarrow \left( \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \right) \text{est}(R) \sqsubseteq \left( \frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$                                                   |

$$S \triangleq F(1, \exists) \alpha, R \sqcap (S^\circ S) \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases



|     |                                                                                                                                    |                                                                                                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right)$                                |                                                                                                                                                            |
| (b) | $= \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right)$                              | $P \left( \frac{1}{\exists} \right) E(\exists) = 1$                                                                                                        |
| (c) | $= \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \frac{1}{\exists} \right) E(\exists)$ | (8.3) $R \sqcap (S^\circ S) \sqsubseteq Q$<br>$\Rightarrow \frac{1}{\exists} \text{est}(R) \frac{1}{\exists} \sqsubseteq \frac{1}{\exists} \text{thin}(Q)$ |
| (d) | $\sqsubseteq \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \right) P(\text{thin}(Q)) E(\exists)$     | $P(\text{thin}(Q)) \text{ union } \sqsubseteq \text{union thin}(Q)$                                                                                        |
| (e) | $\sqsubseteq \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \right) E(\exists) \text{thin}(Q)$        | $\frac{V}{\exists} = \frac{V}{\exists} P \left( \frac{1}{\exists} \right) \text{ union}$                                                                   |
| (f) | $= \frac{F(\exists, 1)}{\exists} \frac{F(1, \exists) \alpha}{\exists} \text{thin}(Q)$                                              | $F(R, 1) F(1, S) = F(R, S)$                                                                                                                                |
| (g) | $= \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$                                                                      |                                                                                                                                                            |

$$= \text{cp}(F) P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}]$$

at the layered network  $(F(A, X) = A + A \times X, B \text{ the paths } V^+, \alpha = [\text{wrap}, \text{cons}]): \frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] - \text{step} \triangleq \text{cpr}$   
 $P(\text{cons}) \text{est}(R)$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} - \frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$$

$$\begin{array}{ccccc} F(A, X) & = & A & + & A \times X \\ & & | & & | \quad | \\ & & R & & R \quad S \\ & & \downarrow & & \downarrow \quad \downarrow \\ F(A', X') & = & A' & + & A' \times X' \end{array}$$

(8.2e)

u  $F(R, S)$  v holds iff one of:

- u, v are both  $\text{inl}$  and  $d \sqsubseteq R \sqsubseteq d'$ : the leaf moves by R;
- u, v are both  $\text{inr}$  and  $p_1 \sqsubseteq R \sqsubseteq q_1 \wedge p_2 \sqsubseteq S \sqsubseteq q_2$ : the vertex moves by R, the rest by S.
- An  $\text{inl}$  never relates to an  $\text{inr}$ :  $R + S \triangleq [R \text{ inl}, S \text{ inr}]$  returns to the summand it came from, and  $F(1) = 1$  allows no crossing.
- $F(\exists, \exists)$  relates  $\text{inr}(\{a, b\}, \{p, q\})$  to  $\text{inr}(a, p), \text{inr}(a, q), \text{inr}(b, p), \text{inr}(b, q)$ , and  $\text{inl}(\{a, b\})$  to  $\text{inl}(a), \text{inl}(b)$ .

## §8.3. Implementing thin

(8.3a)

setify :  $[A] \rightarrow PA$ ,  $\text{cup} : PA \times PA \rightarrow PA$ ,  $\text{cp}(F) \text{ cp}(F) \triangleq \frac{F(\exists)}{3}$ ,  $\text{listcp} : F(L) \rightarrow LF$ ,  $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$  for  $\leq$  a connected preorder.

$\text{thinlist}(Q)$  is any  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$  with  $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$ ; one is  $Q[\text{nil}, \text{bump } Q]$ ,  $\text{bump } Q (a, []) = [a]$ ,  $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$ .

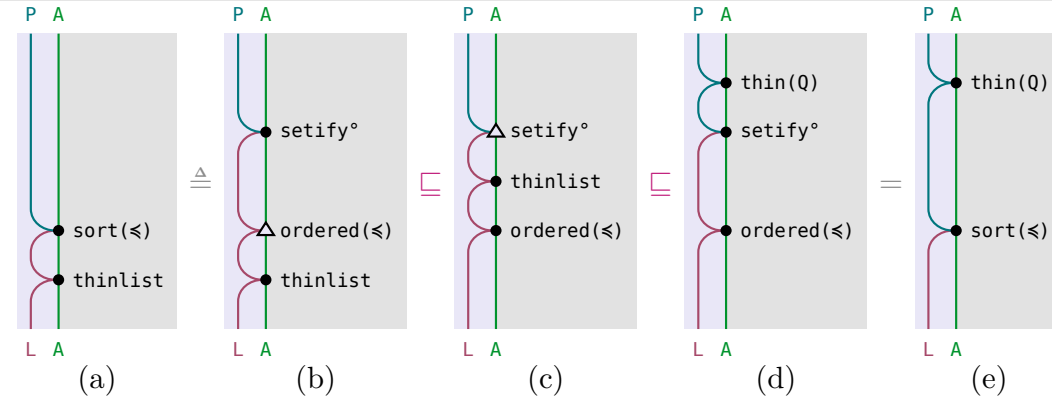
**Binary thinning** data:  $S = (f_1 p_1) \cup (f_2 p_2)$  with  $p_1, p_2$  coreflexive;  $Q$  a preorder with  $Q \sqsubseteq R$  and both  $f_1 p_1, f_2 p_2$  monotonic on  $Q$ ;  $\leq$  a connected preorder with both  $f_1, f_2$  monotonic on  $\leq$ ;  $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$ .

(8.3b)

| the law                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | what it says                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1 $\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$<br>(8.5), $Q$ connected, $xs$ non-empty                                                                                                                                                                                                                                                                                                                                                                            | what thinning should come to when it can: one element                                                              |
| 2 $\text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ (8.6) — (8.6)<br>$\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sort}(\leq) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ | thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys |
| 3 $\text{sort}(\leq) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)                                                                                                                                                                                                                                                                                                                                                                                                              | a minimum of the sorted list is a minimum of the set                                                               |
| 4 $\text{sort}(f \leq f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(\leq)$ (8.8)                                                                                                                                                                                                                                                                                                                                                                                             | shunt a function through a sort                                                                                    |
| 5 $\text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$ (8.9), $p$ coreflexive                                                                                                                                                                                                                                                                                                                                                                                    | filtering a sorted list sorts the restricted set                                                                   |
| 6 $(\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)                                                                                                                                                                                                                                                                                                                                                                         | merging two sorted lists sorts their union                                                                         |
| 7 $F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$<br>(8.11), $F$ linear                                                                                                                                                                                                                                                                                                                                                                          | $\text{listcp}$ is the list implementation of the cartesian product $\text{cp}(F)$                                 |

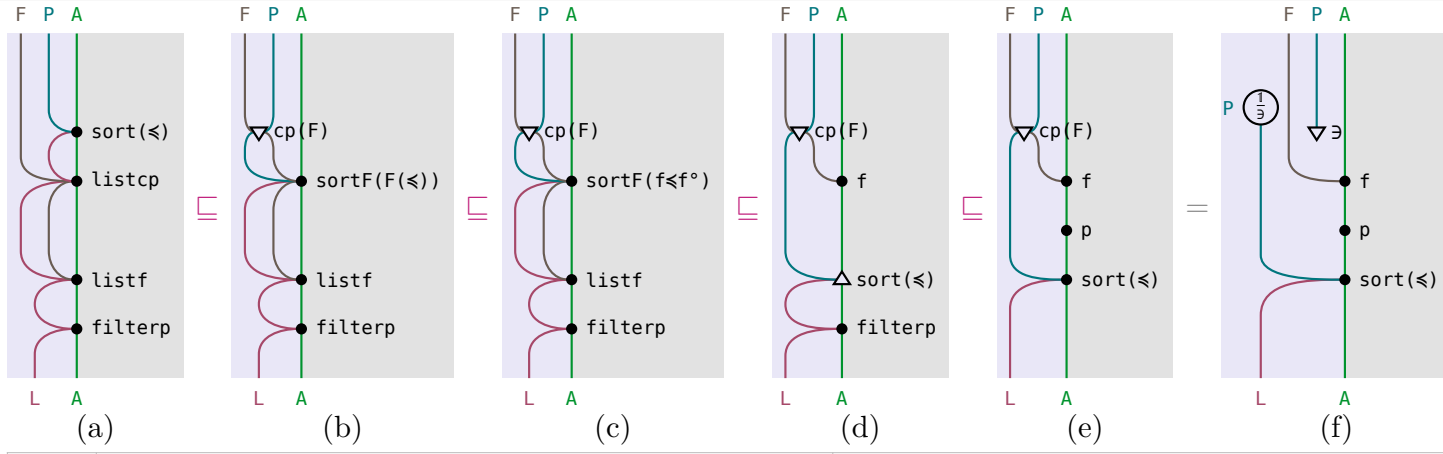
**Map(setify)  $\wedge$  Coreflexive(ordered( $\leq$ ))  $\wedge$  thinlist  $\sqsubseteq$  subseq  $\wedge$  ordered( $\leq$ ) subseq  $\sqsubseteq$  subseq ordered( $\leq$ )  $\wedge$  thinlist setify  $\sqsubseteq$  setify thin( $Q$ )  $\Rightarrow$  sort( $\leq$ ) thinlist  $\sqsubseteq$  thin( $Q$ ) sort( $\leq$ )**  
a thinning of the sorted list lists a thinning of the set —  $\leq$  a connected preorder,  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ .

(8.6)



|     |                                                                         |                                                                                                                                                                                                                                                                                  |
|-----|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{sort}(\leq) \text{ thinlist}$                                    | $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$                                                                                                                                                                                                         |
| (b) | $\triangleq \text{setify}^\circ \text{ ordered}(\leq) \text{ thinlist}$ | $\text{Coreflexive}(\text{ordered}(\leq)) \wedge g \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \Rightarrow \text{setify}^\circ \text{ ordered}(\leq) g \sqsubseteq \text{setify}^\circ g \text{ ordered}(\leq)$ |
| (c) | $\sqsubseteq \text{setify}^\circ \text{ thinlist ordered}(\leq)$        | $\text{Map}(\text{setify}) \wedge g \text{ setify} \sqsubseteq \text{setify } T \Rightarrow \text{setify}^\circ g \text{ ordered}(\leq) \sqsubseteq T \text{ setify}^\circ \text{ ordered}(\leq)$                                                                                |
| (d) | $\sqsubseteq \text{thin}(Q) \text{ setify}^\circ \text{ ordered}(\leq)$ |                                                                                                                                                                                                                                                                                  |
| (e) | $= \text{thin}(Q) \text{ sort}(\leq)$                                   |                                                                                                                                                                                                                                                                                  |

**Map(f)  $\wedge$  ( $\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$ )  $\wedge$  MonoAlg(f,  $\leq$ )  $\wedge$  sortF(f  $\leq$  f°) listf  $\sqsubseteq$  Lemma 8.1**  
**P(f) sort( $\leq$ )  $\wedge$  sort( $\leq$ ) filterp  $\sqsubseteq$  E(p) sort( $\leq$ )  $\wedge$  F(sort( $\leq$ )) listcp  $\sqsubseteq$  cp(F) sortF(F( $\leq$ ))**  
 **$\Rightarrow$  F(sort( $\leq$ )) listcp listf filterp  $\sqsubseteq$   $\frac{F(\exists) \text{ f p}}{\exists}$  sort( $\leq$ )**  
 one sorted list built from sorted arguments, instead of a set built and then sorted — f : FA→A monotonic  
 on  $\leq$ , p coreflexive, F linear.



|     |                                                           |                                                                                                 |
|-----|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| (a) | F(sort( $\leq$ )) listcp listf filterp                    | F(sort( $\leq$ )) listcp $\sqsubseteq$ cp(F) sortF(F( $\leq$ ))                                 |
| (b) | $\sqsubseteq$ cp(F) sortF(F( $\leq$ )) listf filterp      | MonoAlg(f, $\leq$ )                                                                             |
| (c) | $\sqsubseteq$ cp(F) sortF(f $\leq$ f°) listf filterp      | sortF(f $\leq$ f°) listf $\sqsubseteq$ P(f) sort( $\leq$ )                                      |
| (d) | $\sqsubseteq$ cp(F) P(f) sort( $\leq$ ) filterp           | sort( $\leq$ ) filterp $\sqsubseteq$ E(p) sort( $\leq$ )<br>Map(f)<br>$\Rightarrow$ P(f) = E(f) |
| (e) | $\sqsubseteq$ cp(F) E(f p) sort( $\leq$ )                 | $\frac{S}{\exists} E(R) = \frac{S \cdot R}{\exists}$                                            |
| (f) | $= \frac{F(\exists) \text{ f p}}{\exists}$ sort( $\leq$ ) |                                                                                                 |

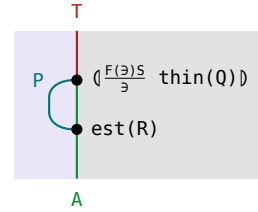
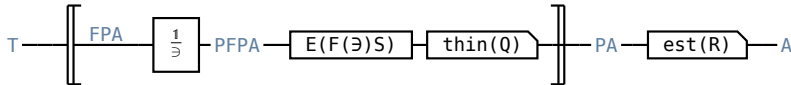
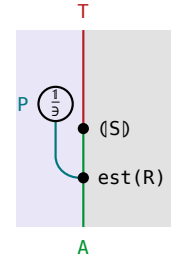
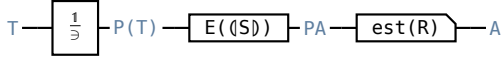
Theorem 8.2

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge \text{MonoAlg}(f_1 \text{ } p_1, Q) \wedge$   
 $\text{MonoAlg}(f_2 \text{ } p_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge$   
 $\text{MonoAlg}(f_2, \leq) \wedge \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq$   
 $P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq)$   
 $\wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq)$   
 $\wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1$   
 $\text{filterp}_1 \wedge g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = (f_1 \text{ } p_1) \cup (f_2 \text{ } p_2) \Rightarrow (\text{listcp} \langle g_1, g_2 \rangle \text{ merge}(\leq))$   
 $\text{thinlist}(Q) \sqsubseteq \text{minlist}(R) \sqsubseteq \frac{(S)}{3} \text{ est}(R)$

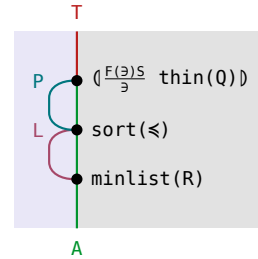
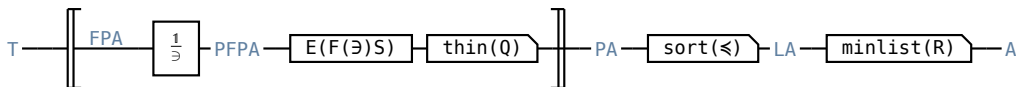
a fold on sorted lists of partial solutions, thinned at every step — at  $\triangleq \text{thinlist}$ 's binary thinning data.

circuit

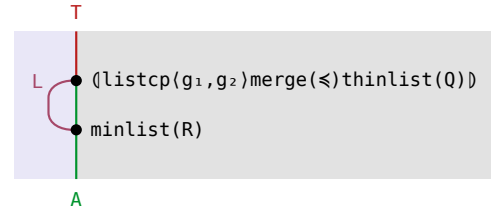
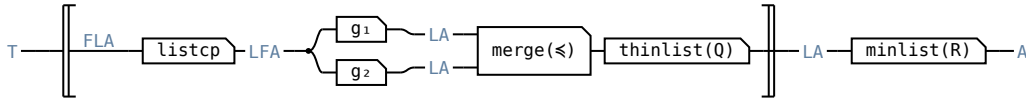
Hinze–Marsden



$\sqsubseteq$  Corollary 8.1  $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge \text{MonoAlg}(S, Q)$   
 $\Rightarrow (\frac{F(3)S}{3} \text{ thin}(Q)) \text{ est}(R) \sqsubseteq \frac{(S)}{3} \text{ est}(R)$  at  $f_1 p_1$  and  $f_2 p_2$  monotonic on  $Q$  —  
 $\triangleq \text{thinlist}$



$\text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R)$  — (8.3b.3) at its  $Q \triangleq R$



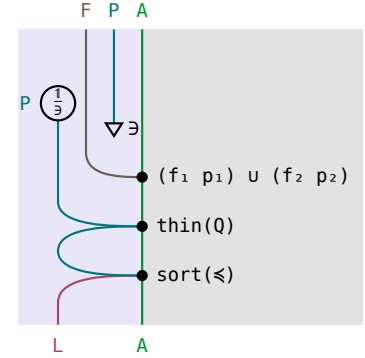
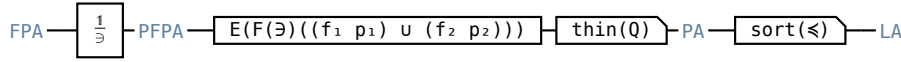
fusion at (8.3c)  $\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge \text{MonoAlg}(f_2, \leq) \wedge \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1$   
 $\sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq)$   
 $\text{filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq) \wedge$   
 $F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq))$   
 $\text{merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$   
 $\Rightarrow F(\text{sort}(\leq)) \text{ listcp} \langle \text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2 \rangle \text{ merge}(\leq)$   
 $\text{thinlist}(Q) \sqsubseteq \frac{F(3) ((f_1 \text{ } p_1) \cup (f_2 \text{ } p_2))}{3} \text{ thin}(Q) \text{ sort}(\leq)$

$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge \text{MonoAlg}(f_2, \leq) \wedge \quad (8.3c) \\
& \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \\
& \text{filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq) \wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \\
& \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \\
& \text{sort}(\leq) \Rightarrow F(\text{sort}(\leq)) \text{ listcp} (\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \text{ merge}(\leq) \text{ thinlist}(Q) \sqsubseteq \\
& \quad \frac{F(\exists) ((f_1 p_1) \cup (f_2 p_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)
\end{aligned}$$

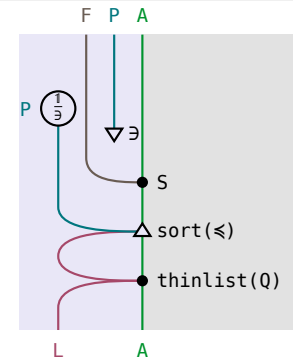
sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit

Hinze–Marsden



$$\begin{aligned}
& \text{FPA} \xrightarrow{\frac{1}{3}} \text{PFPA} \xrightarrow{E(F(\exists)S)} \text{PA} \xrightarrow{\text{sort}(\leq)} \text{thinlist}(Q) \xrightarrow{\text{LA}} \\
& \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \quad \text{--- (8.3b.2)}
\end{aligned}$$



$$\begin{aligned}
& \text{FPA} \xrightarrow{\frac{1}{3}} \text{PFPA} \xrightarrow{E(F(\exists)f_1 p_1)} \text{PA} \xrightarrow{\text{cup}} \text{PA} \xrightarrow{\text{sort}(\leq)} \text{thinlist}(Q) \xrightarrow{\text{LA}} \\
& = \left( \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \right) \text{ cup } \text{sort}(\leq) \text{ thinlist}(Q) \\
& S = (f_1 p_1) \cup (f_2 p_2) \quad \text{--- } \triangleq \text{ thinlist, then } \triangleq \text{ cup}
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \xrightarrow{\frac{1}{3}} \text{PFPA} \xrightarrow{E(F(\exists)f_1 p_1)} \text{PA} \xrightarrow{\text{sort}(\leq)} \text{merge}(\leq) \xrightarrow{\text{thinlist}(Q)} \text{LA} \\
& = \left( \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \right) (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \text{ thinlist}(Q) \\
& (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup } \text{sort}(\leq) \quad \text{--- (8.3b.6)}
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \xrightarrow{\frac{1}{3}} \text{PFPA} \xrightarrow{E(F(\exists)f_1 p_1)} \text{PA} \xrightarrow{\text{sort}(\leq)} \text{LA} \xrightarrow{\text{merge}(\leq)} \text{thinlist}(Q) \xrightarrow{\text{LA}} \\
& = \left( \frac{F(\exists)f_1 p_1}{\exists} \text{ sort}(\leq), \frac{F(\exists)f_2 p_2}{\exists} \text{ sort}(\leq) \right) \text{ merge}(\leq) \text{ thinlist}(Q) \\
& (X, Y) (\text{sort}(\leq) \times \text{sort}(\leq)) = (X \text{ sort}(\leq), Y \text{ sort}(\leq)) \quad \text{--- absorption}
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \xrightarrow{F(\text{sort}(\leq))} \text{FLA} \xrightarrow{\text{listcp}} \text{LFA} \xrightarrow{\text{listf}_1 \text{ filterp}_1} \text{LA} \xrightarrow{\text{listf}_2 \text{ filterp}_2} \text{LA} \xrightarrow{\text{merge}(\leq)} \text{thinlist}(Q) \xrightarrow{\text{LA}} \\
& F(\text{sort}(\leq)) \text{ listcp} (g_1, g_2) \text{ merge}(\leq) \text{ thinlist}(Q) \\
& \text{Lemma 8.1 } \text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, \leq) \wedge \\
& \text{sortF}(f \leq f^\circ) \text{ listf} \sqsubseteq P(f) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp} \sqsubseteq E(p) \text{ sort}(\leq) \wedge \\
& F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \\
& \Rightarrow F(\text{sort}(\leq)) \text{ listcp } \text{listf } \text{filterp} \sqsubseteq \frac{F(\exists) f p}{\exists} \text{ sort}(\leq) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then} \\
& X(g_1, g_2) \sqsubseteq (Xg_1, Xg_2) \quad \text{--- } X \langle R, S \rangle \sqsubseteq \langle X R, X S \rangle ; g_i \triangleq \text{list}(f_i) \text{ filter}(p_i) \quad \text{--- } \triangleq \\
& \text{thinlist}
\end{aligned}$$

## §8.4. The knapsack problem

(8.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}$ ,  $\text{value} \triangleq \text{list}(\text{vol}) \text{ sum}$ ,  $\text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}$ .

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2])$ , within  $w$  the coreflexive on  $\text{xs}$  with weight  $\text{xs} \leq w$ ,  $0 \leq w$ .

$R \triangleq \text{value} \geq \text{value}^\circ$ ,  $Q \triangleq \text{Rn}(\text{weight} \leq \text{weight}^\circ)$ ,  $P \triangleq R, .$

$\text{FA} = 1 + \text{Item} \times A$ ,  $\text{listcp} = \text{wrap} + \text{cpr}$ ,  $g_1 \triangleq \text{list}([\text{nil}, \text{cons}])$   $\text{filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$ ,  
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2]$ .

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w)$ ,  $h_2 \triangleq \text{list}(\pi_2)$ .

(8.4b)

| the law                                                                                                                                        | what it says                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 1 $(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$<br>FALSE                                      | not monotonic on $R$ : a selection of greater value need not still fit once one more item goes in |
| 2 $(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$<br>$(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$ , | both halves are monotonic on $Q$ once ties in value are broken by weight                          |

$0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{nil}, \text{cons}] R$  (8.4c)  
 $[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sort}(R) \wedge \text{sortF}([\text{nil}, \pi_2] R [\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq$   
 $P([\text{nil}, \pi_2]) \text{sort}(R) \wedge \text{sort}(R) \text{filterp}_1 \sqsubseteq E(\text{within}(w)) \text{sort}(R) \wedge F(\text{sort}(R)) \text{listcp} \sqsubseteq \text{cp}(F)$   
 $\text{sortF}(F(R)) \wedge (\text{sort}(R) \times \text{sort}(R)) \text{merge}(R) \sqsubseteq \text{cup } \text{sort}(R) \wedge \text{sort}(R) \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt}))$   
 $\text{sort}(R) \wedge \text{sort}(R) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp } (\text{listf}_1 \text{filterp}_1, \text{listf}_2) \text{merge}(R) \text{thinlist})$   
 $\text{minlist} \sqsubseteq \frac{\text{subseq } \text{within}(w)}{\supset} \text{est}(R)$

the knapsack problem, as a fold that thins the packings kept at each item

| circuit                                                                                                                                                                                                                                                                                                                                         | Hinze–Marsden |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <p><math>\text{subseq } (\text{within } w) \text{ est}(R)</math></p>                                                                                                                                                                                                                                                                            |               |
| <p><math>\frac{[([\text{nil}, \text{cons}] \text{within } w) \cup [\text{nil}, \pi_2]]}{\supset} \text{est}(R)</math></p> <p>fusion, weights non-negative.</p>                                                                                                                                                                                  |               |
| <p><math>([\text{listcp } (g_1, g_2) \text{merge } R \text{thinlist}(Q)] \text{minlist}(R)</math></p> <p>binary thinning theorem, at <math>P \triangleq R</math>, <math>F</math> linear, <math>Q</math> from (8.4b.2)</p>                                                                                                                       |               |
| <p><math>([\text{nil}, \text{cpr } (h_1, h_2) \text{merge } R \text{thinlist}(Q)] \text{minlist}(R)</math></p> <p><math>= \text{listcp} = \text{wrap} + \text{cpr}</math>, <math>g_i = [\text{list}(\text{nil}), h_i] \triangleq \text{within}</math>;<br/> <math>\text{minlist}(R)</math> is head, packings coming out in descending value</p> |               |

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.

new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g₁≐list([wrap wrap,new]), g₂≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h₁≐list(new),
h₂≐list(glue) filter(ok w).

```

| the law                                                                                                                                                                                             | what it says                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| $(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$                                                                                                                           | <b>glue</b> is not monotonic on $R$ : the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |
| $(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$<br>$(1 \times Q) (\text{glue } (ok \ w)) \sqsubseteq \text{glue } (ok \ w) Q$<br>$\text{cons monotonic on collect} \leq \text{collect}^\circ.$ | both halves are monotonic on $Q$ once ties in waste are broken by the first line                                                   |
| merge $\tau = \text{cat}$ ; $P \triangleq \text{head prefix head}^\circ$ also serves                                                                                                                | $\tau$ needs no sorting at all, and <b>prefix</b> is a linear order on first lines of paragraphs of one input                      |

$$\begin{aligned}
& (\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \subseteq Y \Rightarrow \text{sortF}(X) \subseteq \text{sortF}(Y)) \wedge \text{sortF}[\text{wrap} \\
& \text{wrap}, \text{new}] \top [\text{wrap wrap}, \text{new}]^\circ) \text{listf}_1 \in P([\text{wrap wrap}, \text{new}]) \text{sort}(\top) \wedge \text{sortF}[\text{wrap wrap}, \text{glue}] \top \\
& [\text{wrap wrap}, \text{glue}]^\circ) \text{listf}_2 \in P([\text{wrap wrap}, \text{glue}]) \text{sort}(\top) \wedge \text{sort}(\top) \text{filterp}_2 \in E(\text{ok}(w)) \text{sort}(\top) \\
& \wedge F(\text{sort}(\top)) \text{listcp} \in \text{cp}(F) \text{sortF}(F(\top)) \wedge (\text{sort}(\top) \times \text{sort}(\top)) \text{merge}(\top) \in \text{cup} \text{sort}(\top) \wedge \\
& \text{sort}(\top) \text{thinlist} \in \text{thin}(Q(\text{len}, w)) \text{sort}(\top) \wedge \text{sort}(\top) \text{minlist} \in \text{est}(R) \Rightarrow (\text{listcp} \\
& (\text{listf}_1, \text{listf}_2 \text{filterp}_2) \text{merge}(\top) \text{thinlist}) \text{minlist} \in \frac{\text{partition fits}(w)}{3} \text{est}(R)
\end{aligned}$$

a paragraph laid out as a fold that thins the layouts kept at each word

Hinze–Marsden

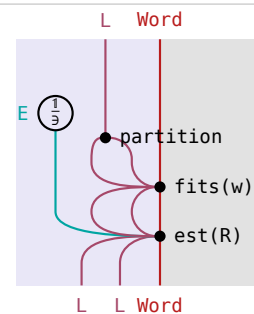


Diagram illustrating the evaluation of a word  $w$  in the context of a stack. The stack contains the word "L Word". The input string is  $([wrap\ wrap,new] u ([wrap\ wrap,glue]ok(w)))$ . The stack is updated with  $est(R)$ .

Diagram illustrating the reduction of a list expression to a single word. The expression is `(listcp(listf1, listf2 filterp2) merge(T) thinlist)`. The diagram shows the expression being reduced to a single word, `minlist`, which is then merged with the rest of the expression to produce the final result, `list`.

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## §8.6. Bitonic tours

(8.6a)

$FA = (City \times City) + (City \times A)$ , the base functor of cons-lists of length at least two;  $listcp = wrap + cpr$ ;  $tc : City \times City \rightarrow Real$ , neither positive nor symmetric.

$start(a, b) = ([a, b], [a, b])$ ,  $dropl(a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$ ,  $dropr(a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$ ,  $tour \triangleq ([start, dropl \cup dropr])$ .

$cost(xs, ys) = outcost\ xs + incost\ ys$ ,  $outcost[a_0, \dots, a_n] = tc(a_0, a_1) + \dots + tc(a_{n-1}, a_n)$ ,  $incost[a_0, \dots, a_n] = tc(a_1, a_0) + \dots + tc(a_n, a_{n-1})$ .

$next \triangleq tail\ head$ ,  $next2 \triangleq next \times next$ ,  $head2 \triangleq head \times head$ ,  $R \triangleq cost \leq cost^\circ$ ,  $Q \triangleq Rn(next2\ next2^\circ) \cap (head2\ head2^\circ)$ ,  $P \triangleq T$ ,  $g_1 \triangleq list([start, dropl])$ ,  $g_2 \triangleq list([start, dropr])$ .

(8.6b)

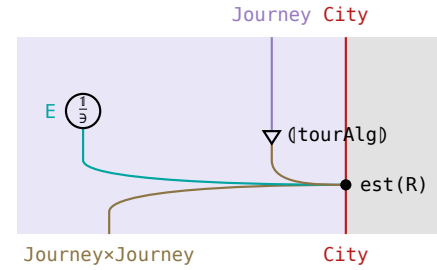
| the law                                                                                                                                        | what it says                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1 $(1 \times R) \text{ dropl} \sqsubseteq \text{dropl}\ R \quad FALSE$<br>$(1 \times R) \text{ dropr} \sqsubseteq \text{dropr}\ R \quad FALSE$ | neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists            |
| 2 $(1 \times Q) \text{ dropl} \sqsubseteq \text{dropl}\ Q$<br>$(1 \times Q) \text{ dropr} \sqsubseteq \text{dropr}\ Q$ ,                       | both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input |

$$\begin{aligned}
 & (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([start, droplFn] \tau [start, droplFn]^\circ) \ g_1 \sqsubseteq \quad (8.6c) \\
 & \quad P([start, droplFn]) \text{ sort}(\tau) \wedge \text{sortF}([start, droprFn] \tau [start, droprFn]^\circ) \ g_2 \sqsubseteq \\
 & \quad P([start, droprFn]) \text{ sort}(\tau) \wedge F(\text{sort}(\tau)) \text{ listcp} \sqsubseteq cp(F) \text{ sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{ cat} \\
 & \quad \sqsubseteq \text{cup} \text{ sort}(\tau) \wedge \text{sort}(\tau) \text{ thinlist} \sqsubseteq \text{thin}(Qc(tc)) \text{ sort}(\tau) \wedge \text{sort}(\tau) \text{ minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp} \\
 & \quad (g_1, g_2) \text{ cat thinlist}] \text{ minlist} \sqsubseteq \frac{(\text{tourAlg})}{3} \text{ est}(R)) \\
 & \quad \text{a least-cost bitonic tour, as a fold that thins the tours kept at each city}
 \end{aligned}$$

circuit

Hinze–Marsden

$$\begin{aligned}
 & dCL(City \times City, City) \xrightarrow{\frac{1}{3}} P(dCL(City \times City, City)) \xrightarrow{E(\text{tourAlg})} P(\text{Journey}(City)^2) \xrightarrow{\text{est}(R)} \text{Journey}(City) \\
 & \text{tour} \xrightarrow{3} \text{est}(R)
 \end{aligned}$$



$$\begin{aligned}
 & dCL(City \times City, City) \xrightarrow{\frac{1}{3}} P(dCL(City \times City, City)) \xrightarrow{E([start, droplFn] \cup [start, droprFn])} P(\text{Journey}(City)^2) \xrightarrow{\text{est}(R)} \text{Journey}(City) \\
 & = \frac{([start, dropl \cup dropr])}{3} \text{est}(R) \\
 & \text{tour} \triangleq ([start, dropl \cup dropr]) \triangleq \text{tour}
 \end{aligned}$$

$$\begin{aligned}
 & dCL(City \times City, City) \xrightarrow{\frac{1}{3}} P(dCL(City \times City, City)) \xrightarrow{E([start, droplFn] \cup [start, droprFn])} P(\text{Journey}(City)^2) \xrightarrow{\text{est}(R)} \text{Journey}(City) \\
 & \sqsubseteq ([\text{listcp}(g_1, g_2) \text{ cat thinlist}(Q)] \text{ minlist}(R)) \\
 & \text{binary thinning theorem, at } P \triangleq T \text{ with merge } \tau = \text{cat}, Q \text{ from (8.6b.2)}
 \end{aligned}$$

$$\begin{aligned}
 & ([start\ wrap, cpr\ (list(dropl), list(dropr)) \text{ cat} \\
 & = \text{thinlist}(Q)] \text{ minlist}(R) \\
 & \text{listcp} = \text{wrap} + \text{cpr}, g_i = [list(start), list(drop_i)] \triangleq \text{tour}; \\
 & \text{quadratic, two tours added per step}
 \end{aligned}$$

